

Conducting Forensic Investigations on System Memory (4e)

Digital Forensics, Investigation, and Response, Fourth Edition - Lab 10

Student:

Rudra Chevdawala

Email:

rudra03@nmsu.edu

Time on Task:

5 hours, 15 minutes

Progress:

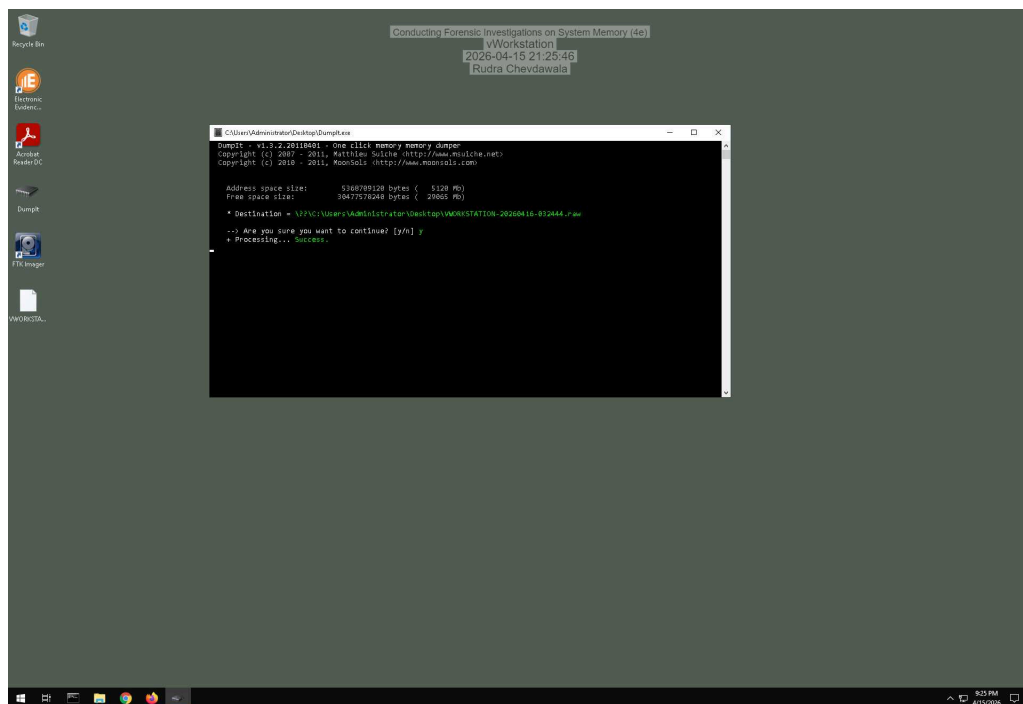
100%

Report Generated: Thursday, April 16, 2026 at 1:10 AM

Section 1: Hands-On Demonstration

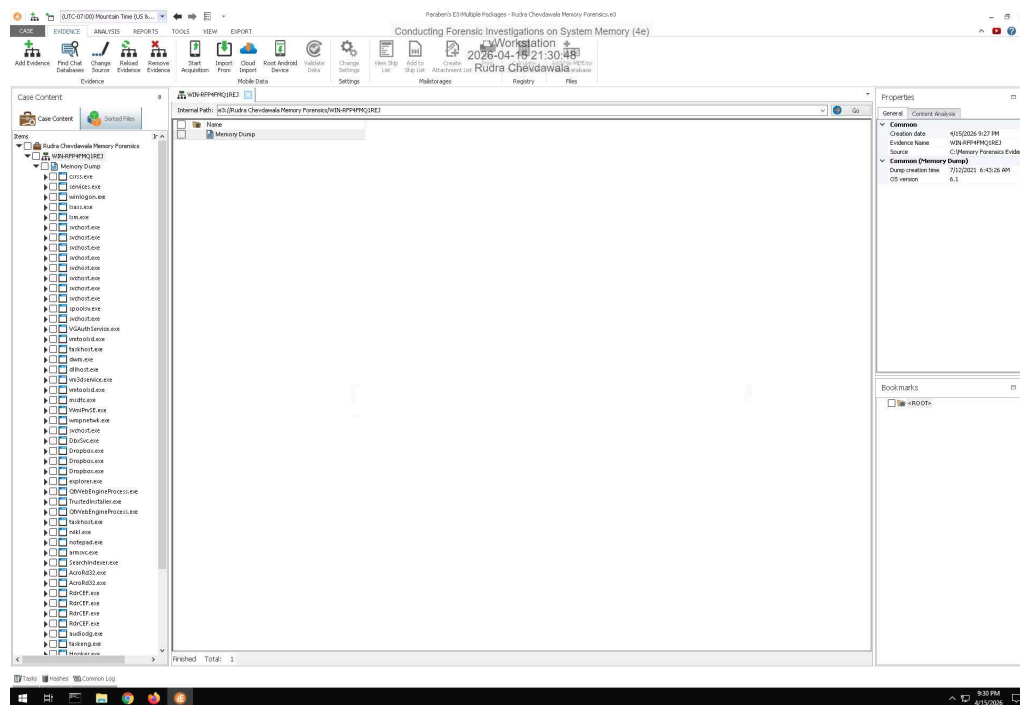
Part 1: Capture Memory using DumpIt

3. Make a screen capture showing the **Dumplt** success notification.



Part 2: Analyze Memory using E3

8. Make a screen capture showing the list of processes in the memory dump.



10. Record the start times for the oldest process and the newest process.

The start time for the oldest process is 7/12/2021 at 4:24:49 AM, and the latest process start time was 7/12/2021 at 6:42:32 AM.

15. Document your findings for the conhost.exe process. What is it and what is it used for?

The conhost.exe process provides essential support for the Command Prompt by enabling it to integrate with Windows Explorer, including features like dragging and dropping files or folders into the terminal window. It stands for "Console Window Host" and is a legitimate Windows system program responsible for managing console window interaction and display. While it is normally safe, attackers may attempt to exploit or imitate it, which can create security vulnerabilities if the process is not genuine.

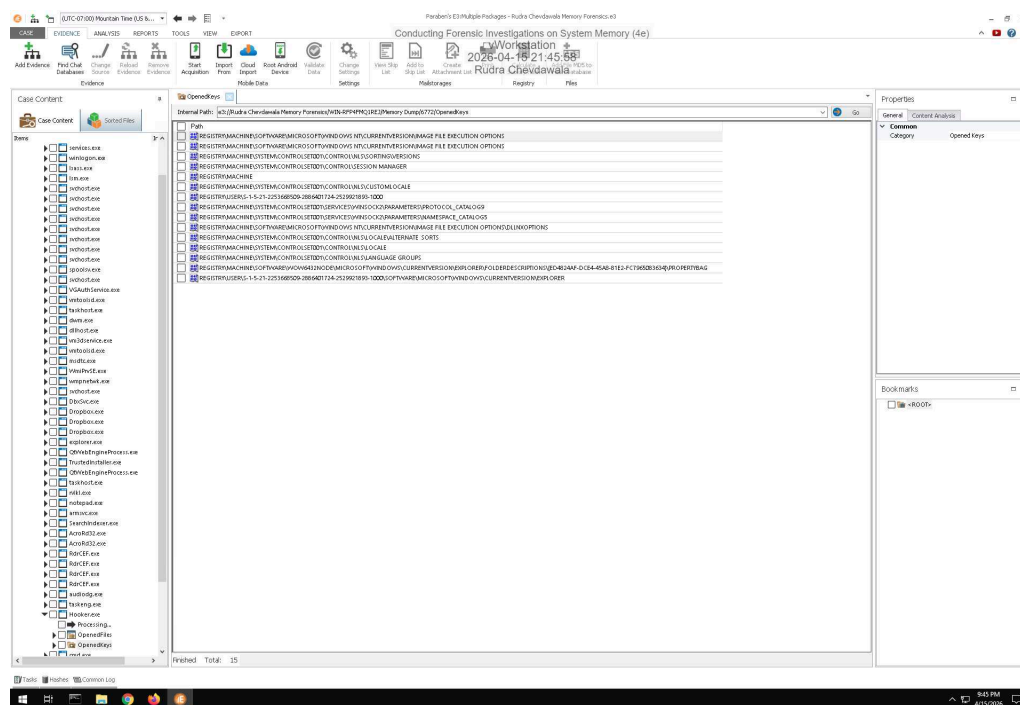
Digital Forensics, Investigation, and Response, Fourth Edition - Lab 10

Digital Forensics, Investigation, and Response, Fourth Edition - Lab 10

17. **Document** your findings for the hooker.exe process. What is it and what is it used for?

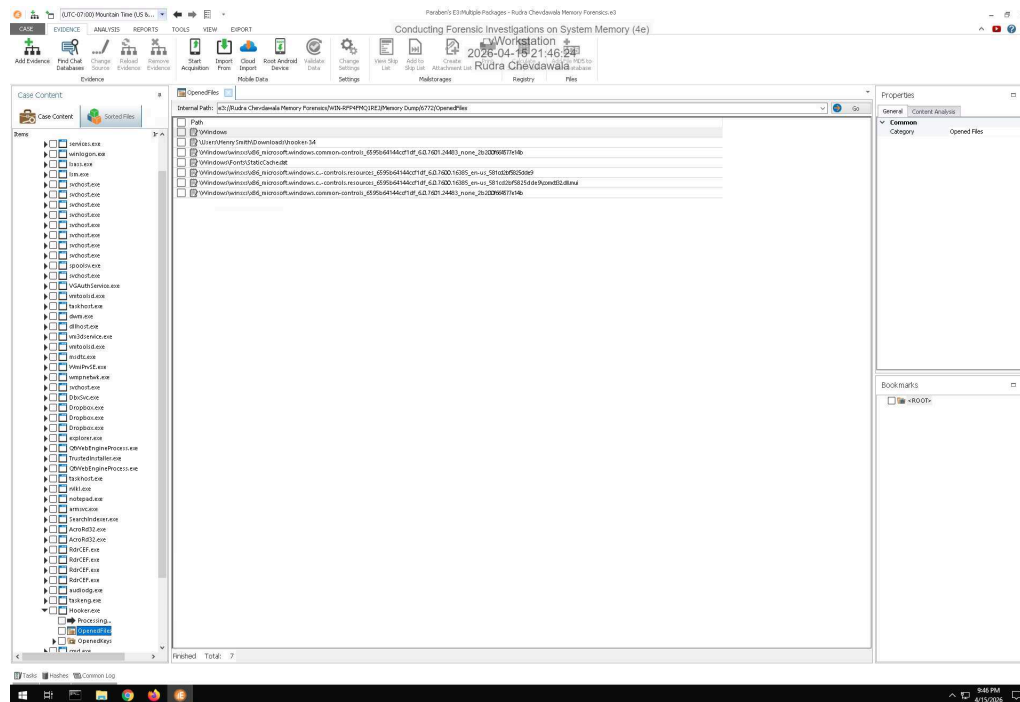
The hooker.exe process is not part of the Windows operating system and is commonly associated with monitoring system activity. It can capture keyboard and mouse input, observe running applications, and communicate over the internet to transmit collected information. Because of this behavior, it is generally considered unsafe and poses significant security risks.

21. **Make a screen capture** showing the **registry keys** opened by the **Hooker.exe** process.



Digital Forensics, Investigation, and Response, Fourth Edition - Lab 10

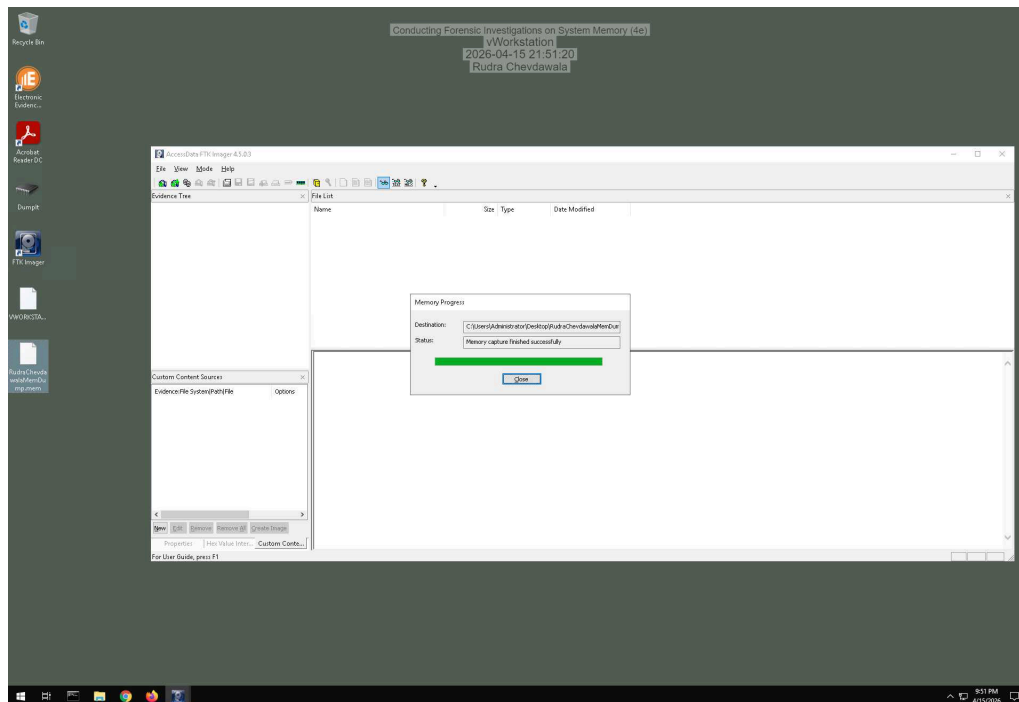
23. **Make a screen capture showing the files opened by the hooker.exe process.**



Section 2: Applied Learning

Part 1: Capture Memory using FTK Imager

6. Make a screen capture showing the *Memory capture finished successfully* confirmation.



Part 2: Analyze Memory using Volatility

7. **Document** your findings for the rvlkl.exe process. What is it and what is it used for?

The rvlkl.exe process is associated with Logixoft's Revealer Keylogger, a monitoring and security utility that tracks user activity on a computer. It records keystrokes, mouse actions, and other inputs, and can generate log files that include screenshots that may be transmitted remotely to the administrator. Although the program is digitally signed and typically runs hidden in the background, it raises privacy and security concerns and is not generally considered fully safe, as it can be misused or behave inconsistently on certain system configurations.

9. **Document** whether any processes are flagged as hidden.

Several processes are flagged as hidden: **services.exe**, **lsass.exe**, **lsim.exe**, **System**, and **csrss.exe**.

Digital Forensics, Investigation, and Response, Fourth Edition - Lab 10

- Yes, the netscan module shows network usage associated with the Hooker.exe or rvlkl.exe processes through Chrome and Dropbox executables.

- The information I found about port 56610 indicates that it is a dynamic, private port commonly used for temporary system activity typically associated with Apple's Xsan Filesystem Access. Ports in this range are often assigned automatically for short-term tasks, such as accessing files or supporting background system functions. In this case, port 56610 appears to be used for routine system operations rather than a fixed or user-defined service.

- [illegible]

Digital Forensics, Investigation, and Response, Fourth Edition - Lab 10

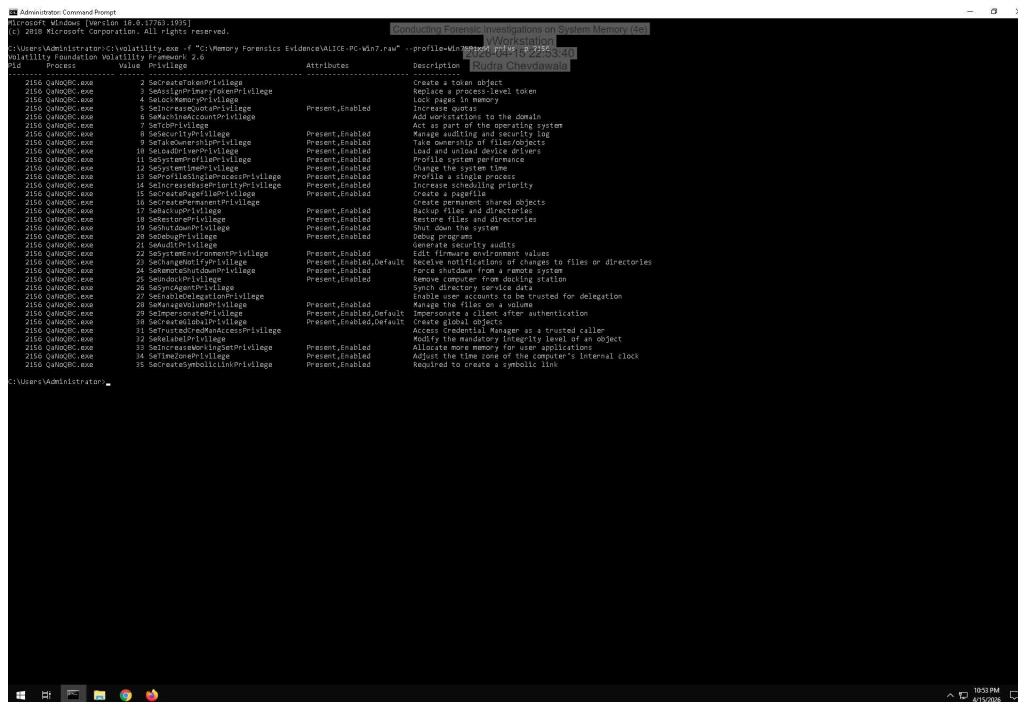
[illegible]

Page 8 of 9

Conducting Forensic Investigations on System Memory (4e)

Digital Forensics, Investigation, and Response, Fourth Edition - Lab 10

Make a screen capture showing the output of your privilege comparison.



```
Administrative Command Prompt
C:\Users\Administrator>systeminfo
[Output of systeminfo command showing system information]
C:\Users\Administrator>
```